

Aly Ahmed

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EDUCATION

University of Waterloo

Waterloo, ON

Bachelor of Applied Science, Mechatronics Engineering, Honours, Artificial Intelligence Option

Apr 2027

- Relevant Courses

- * Autonomous Mobile Robots, Fundamentals of Computational Intelligence, Control Applications, Introduction to Computer Structures and Real-Time Systems, Microprocessor Systems and Interfacing for Mechatronics Engineering

EXPERIENCE

Robotics Software Engineer

Burnaby, BC

[Red Rabbit Robotics](#)

Jan 2026 – Aug 2026

- Key member of a small team that built a robot from scratch, developing control, vision, teleoperation, and hardware; the robot was deployed in front of customers across the U.S.
- Engineered a C++/Python and ROS2 SDK for QDD actuators using SocketCAN to enable arm control
- Built a robotic arm PID auto-tuner using Bayesian Optimization (Optuna) to search for controller gains automatically
- Developed a vision-to-placement pipeline mapping 2D image coordinates to 3D points with Orbbec RGB-D data, and a ChArUco-based camera-to-robot calibration routine
- Implemented teleoperation over UDP and TCP/IP, and ran teleoperation sessions to collect multi-modal training data for VLA models
- Benchmarked hardware running inference, camera streaming, and control algorithms to select the robot's compute platform
- Stress-tested the robot across tasks and conditions, feeding issues back into software; assembled and wired robotic arms and control boxes, including the emergency stop

Robotics Perception and Control R&D

Waterloo, ON

[Robotic Interaction, Perception and Learning lab – University of Waterloo](#)

May 2025 – August 2025

- Designed a probabilistic estimation pipeline for modeling and predicting the motion of flexible objects to support robot manipulation, implemented with factor graphs (GTSAM), ROS2, and Eigen. (C++)
- Implemented cartesian and joint impedance control with time-optimal trajectory tracking on the Franka Emika Panda, utilizing ROS2-Control (C++, Python)

Robotics Research Assistant

Waterloo, ON

[Active & Interactive Robotics Lab – University of Waterloo](#)

Sep 2025 – Dec 2025

- Explored methods for implementing impedance control on the TIAGo Pro from Pal Robotics

Platform Software Developer

Kanata, ON

[Ford Motor Company](#)

Sep 2024 – Dec 2024

- Executed embedded cryptography software improvements, successfully resolving 1000+ code issues and developed unit tests utilizing Google Test (C++)

Software Test Developer

Kanata, ON

[Blackberry QNX](#)

Jan 2024 – Apr 2024

- Developed automated tests, and performed manual tests on Qualcomm, Bosch, and NXP ECUs, running QNX, and android automotive, producing invaluable engineering data, facilitating successful OEM demos
- Supported proof of concept demos to major BlackBerry customers like Nissan, and Ford through integration, development, and testing

SKILLS

Languages: C, C++, Python, Bash

Frameworks & OS: ROS/ROS2, FreeRTOS, QNX, Linux, OpenCV, GTSAM

Tools & Architecture: Git, Docker, AWS, Networking (TCP, UDP, CAN, SPI, I2C, UART)

Hardware & CAD: Altium Designer, SolidWorks, Fusion 360, Oscilloscopes, Soldering, Wire Harnessing